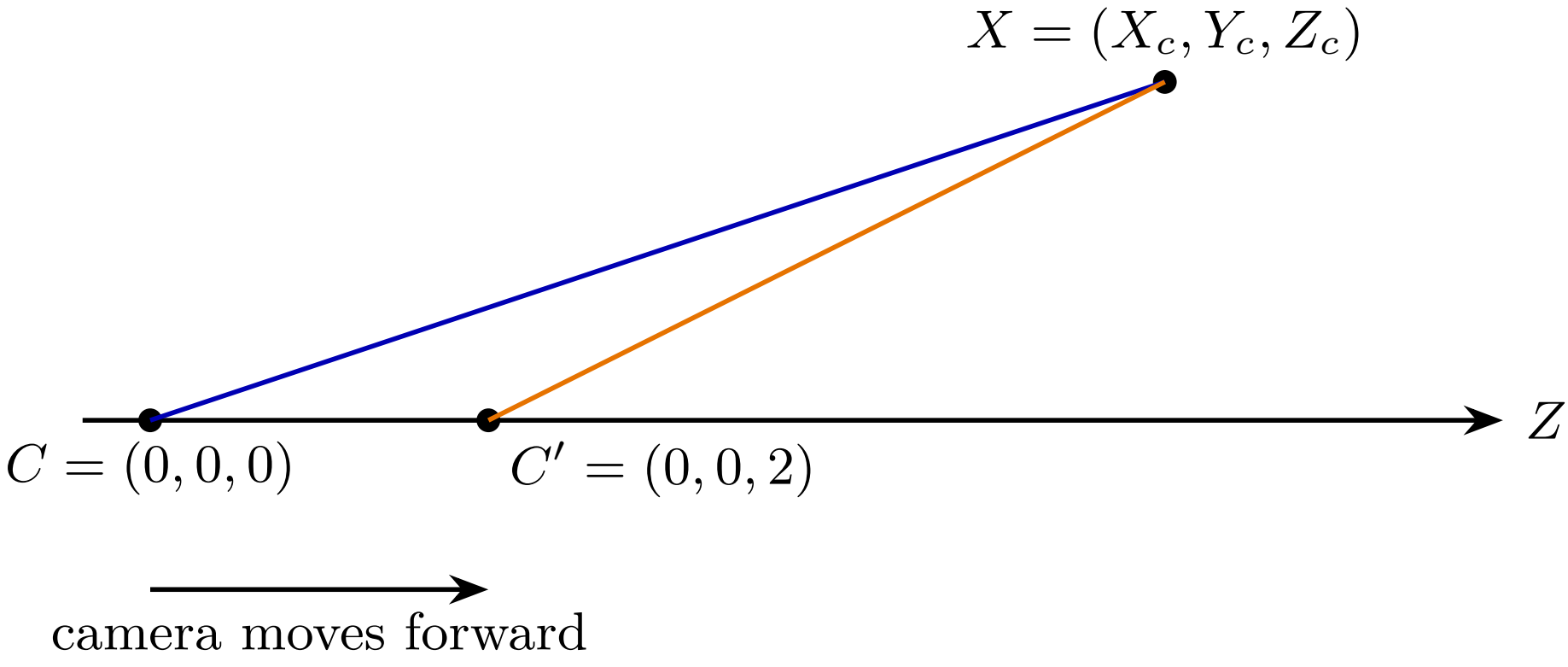The point stays fixed. Its camera-relative depth changes from Z_c to $Z_c - 2$.